

Jonathan Viquerat

R&D engineer

PhD in applied mathematics

06130 Grasse

✉ jonathanviquerat@gmail.com

🌐 [Webpage](#)

Professional experience

- 2025 – Present **R&D engineer**, Ansys/Synopsys, (Sophia Antipolis, France), Lumerical team, Numerical photonics and optimization
- 2018 – 2025 **Research engineer**, CEMEF Mines Paris PSL, (Sophia Antipolis, France), CFL research group, Coupling of numerical methods with machine learning
- 2015 – 2018 **Research engineer**, INRIA, (Sophia Antipolis, France), Nachos project-team, Development of a high-order discontinuous Galerkin solver suite for nano-optics problems
- 2012 – 2015 **PhD in applied mathematics and numerical simulation**, INRIA, (Sophia Antipolis, France), Nachos project-team, Discontinuous Galerkin time-domain method for nanophotonics

Research axes, projects and developments

- drl-based flow control** Coupling of CFD environments with deep reinforcement learning agents for flow control, including :
- ♦ Two reviews ([paper](#) and [paper](#)),
 - ♦ Methodological contributions on distributed environments and state representation learning ([paper](#)),
 - ♦ The co-advising of a post-doctoral student ([A. Batteni](#)),
 - ♦ DRAGONFLY, a modular DRL library ([github](#)),
 - ♦ BEACON, a set of flow control benchmarks ([github](#))
- gradient-free optimization** Multiple works on optimization methods, mainly in the scope of fluid dynamics, including :
- ♦ The PBO (policy-based optimization) method, a deep reinforcement learning-based approach to optimization ([paper](#) and [github](#)),
 - ♦ Multiple applicative papers (see [selected publications](#)),
 - ♦ SPARKLE, a modular gradient-free optimization library ([github](#))
- exalib** A modern C++ CFD library, developed with A. Larcher, including :
- ♦ A modular parallel finite element framework,
 - ♦ A set of compressible and incompressible flow solvers,
 - ♦ A set of mesh modification tools,
 - ♦ A continuous integration framework.
- supervised ML for CFD** Several works on neural networks-based CFD surrogates, including :
- ♦ Convolutional neural networks models for laminar and turbulent flows (see the MINDS project below),
 - ♦ Autoregressive transformer-based graph neural networks approaches
 - ♦ The co-advising of three PhD students ([P. Garnier](#) (ERC Cure), [A. Bazzi](#), and [X.M.V. Nguyen](#))
 - ♦ Multiple methodological papers (see [selected publications](#)),
- minds** A CARNOT project led by E. Hachem, bringing together numerical computation and data sciences, including :
- ♦ A research activity organized around supervised and reinforcement learning for CFD problems,
 - ♦ The co-advising of three PhD students ([J. Chen](#), [H. Ghraieb](#), [G. el Haber](#) and [A. Patil](#)),
 - ♦ A website (see <https://cfl-minds.github.io/>),
 - ♦ Regular seminars, given by experts in the field.

- diogenes** A discontinuous Galerkin library for electromagnetics and nano-optics applications, written in modern Fortran, including :
- ◇ A parallel discontinuous Galerkin time-domain solver for Maxwell's equations, with an advanced mesh processing library, a mesh partitioning tool, a material processing tool for metal and semiconductor permittivity laws, and a coupling with optimization libraries,
 - ◇ The computation of a large set of real-life problems, set up in collaboration with academic and industrial partners (CNRS LPMC, Bristol University, C2N, CEA LETI, ...),
 - ◇ A website (see <https://diogenes.inria.fr/>).

Studies

- 2012 **Master's degree internship (2nd year)**, INRIA, (Sophia Antipolis, France), Nachos project-team, Topic : Discontinuous Galerkin time-domain method for nanophotonics
Reference : Stéphane Lanteri (stephane.lanteri@inria.fr)
- 2011 **Master's degree internship (1st year)**, University College, (London, UK), Mechanical engineering department, Topic : Assessment of transcatheter aortic valve devices by numerical simulation on commercial solver
Reference : Gaetano Burriesci, g.burriesci@ucl.ac.uk
- 2009 – 2012 **Engineering degree in applied mathematics**, ENSTA ParisTech (Paris, France)
- 2011 – 2012 **Master's degree in modelisation and numerical simulation, with distinctions**, CEA (Saclay, France)

Selected publications ([link to full list](#))

- 2026 P. Garnier, V. Lannelongue, J. Viquerat, E. Hachem *Training transformers for mesh-based simulations*, **Neural Computing and Applications**, vol. 38, iss. 11, pp. 460
- 2025 P. Garnier, V. Lannelongue, J. Viquerat, E. Hachem *MeshMask : Physics-based simulations with masked graph neural networks*, accepted in **ICLR 2025**
- 2024 J. Viquerat, P. Meliga, P. Jeken-Rico, E. Hachem, *Beacon, a lightweight deep reinforcement learning benchmark library for flow control*, **Applied Sciences**, vol. 14, iss. 9, pp. 3561
- 2023 A. Patil, J. Viquerat, E. Hachem, *Autoregressive transformers for data-driven spatiotemporal learning of turbulent flows*, **APL Machine Learning**, vol. 1, iss. 4, pp. 046101
- 2023 J. Viquerat, E. Hachem, *Parallel bootstrap-based on-policy deep reinforcement learning for continuous fluid flow control applications*, **Fluids**, vol. 8, iss. 7, pp. 304
- 2023 J. Viquerat, R. Duvigneau, P. Meliga, A. Kuhnle, E. Hachem, *Policy-based optimization : single-step policy gradient method seen as an evolution strategy*, **Neural Computing and Applications**, vol. 35, iss. 1, pp. 449 – 467
- 2022 J. Viquerat, P. Meliga, A. Larcher, E. Hachem, *A review on deep reinforcement learning for fluid mechanics : an update*, **Physics of Fluids**, vol. 34, iss. 11, pp. 111301
- 2022 J. Chen, J. Viquerat, E. Hachem, *A twin-decoder structure for incompressible laminar flow reconstruction with uncertainty estimation around 2D obstacles*, **Neural Computing and Applications**, vol. 34, pp. 6289 – 6305
- 2021 A. Patil, J. Viquerat, E. Hachem, *Robust deep learning for emulating turbulent viscosities*, **Physics of Fluids**, vol. 33, iss. 10, pp. 105118
- 2021 J. Viquerat, J. Rabault, A. Kuhnle, H. Ghraieb, E. Hachem, *Direct shape optimization through deep reinforcement learning*, **Journal of Computational Physics**, vol. 428, pp. 110080
- 2021 P. Garnier, J. Viquerat, J. Rabault, A. Larcher, A. Kuhnle, E. Hachem, *A review on deep reinforcement learning for fluid mechanics*, **Computer and Fluids**, vol. 225, pp. 104973
- 2019 V. Belus, J. Rabault, J. Viquerat, Z. Che, E. Hachem, U. Reglade, *Exploiting locality and translational invariance to design effective deep reinforcement learning control of the 1-dimensional unstable falling liquid film*, **AIP Advances**, vol. 9, pp. 125014
- 2019 J. Viquerat, N. Schmitt, C. Scheid, *Simulating 3D periodic structures at oblique incidences with discontinuous Galerkin time-domain methods : theoretical and practical considerations*, **SMAI Journal of Computational Mathematics**, vol. 5, pp.131 – 159
- 2019 J. Viquerat, *Efficient time-domain numerical analysis of waveguides with tailored wideband pulses*, **Microwave and Optical Technology Letters**, vol. 61, pp. 1534 – 1539

- 2018 J. Viquerat, *Fitting experimental dispersion data with a simulated annealing method for nano-optics applications*, **Journal of Nanophotonics**, vol. 12, pp. 036014
- 2018 N. Schmitt, C. Scheid, J. Viquerat, S. Lanteri, *Simulation of three-dimensional nanoscale light interaction with spatially dispersive metals using a high-order curvilinear DGTD method*, **Journal of Computational Physics**, vol. 373, pp. 210 – 229
- 2016 J. Viquerat, S. Lanteri, *Simulation of near-field plasmonic interactions with a local approximation order discontinuous Galerkin time-domain method*, **Photonics and Nanostructures-Fundamentals and Applications**, vol. 18, pp. 43 – 58
- 2016 N. Schmitt, C. Scheid, S. Lanteri, A. Moreau, J. Viquerat, *A DGTD method for the numerical modeling of the interaction of light with nanometer scale metallic structures taking into account non-local dispersion effects*, **Journal of Computational Physics**, vol. 316, pp. 396 – 415
- 2015 J. Viquerat, *Simulation of electromagnetic wave propagation in nano-optics with a high-order discontinuous Galerkin time-domain method*, PhD thesis (see <https://www.archives-ouvertes.fr/tel-01272010/>)
- 2015 J. Viquerat, C. Scheid, *A 3D curvilinear discontinuous Galerkin time-domain solver for nanoscale light-matter interactions*, **Journal of Computational and Applied Mathematics**, vol. 289, pp. 37 – 50

Conferences

- 2022 AI4Waves, Sophia-Antipolis (France), *DRL for flow control*
- 2018 GDR Ondes, Paris (France), *Diogenes : a DG-based software suite for nano-optics problems*
- 2014 Acomen, Ghent (Belgium), *A curvilinear discontinuous Galerkin time-domain method for nanophotonics*
- 2014 Meta, Singapore, *Discontinuous Galerkin time-domain method for nanophotonics*
- 2013 Waves, Tunis (Tunisia), *Discontinuous Galerkin Time-Domain method for nanophotonics*
- 2012 GDR Ondes, Troyes (France), *Méthode Galerkin discontinue en domaine temporel pour la propagation d'ondes électromagnétiques en nano-optique*

Teaching and supervising

- 2018 – Present **Co-advisor of 7 PhD students and 1 post-doctoral student**, CEMEF Mines ParisTech (Sophia Antipolis, France)
- 2019 – Present **Advisor of 7 M.Sc. interns**, CEMEF Mines ParisTech (Sophia Antipolis, France)
- 2010 – 2011 **Computing science teacher in "Classes préparatoires"**, Lycée Marcelin Berthelot (Saint-Maur des Fossés, France)

Skills

Programming	C++, Python, Fortran
HPC	MPI, OpenMP
Machine learning	Tensorflow, Pytorch, JAX
Meshes	Gmsh, MeshGems
Post-treatment	Paraview, Vizir, Medit
Development	Git, CI, CMake
Systems	Linux, OsX
Misc.	LaTeX, HTML, CSS
Languages	English (fluent)

Hobbies

- Sports** Climbing, alpinism, ski-touring, hiking
- Others** Photography